

H524 Introduction to Biostatistics

Week 7 Lab: Nonparametric Methods - ANSWER KEY

Fall 2025

1 Exercise 1: Shapiro-Wilk Test

```
# Generate example data
set.seed(524)
normal_data <- rnorm(30, mean=100, sd=15)
skewed_data <- rexp(30, rate=0.02)

# Check normality
shapiro.test(normal_data)
shapiro.test(skewed_data)
```

Answers:

1. P-value for `normal_data`: 0.2371
2. P-value for `skewed_data`: 0.0025
3. For `normal_data`, use **parametric** tests ($p = 0.2371 \geq 0.05$, normality assumption OK)
4. For `skewed_data`, use **nonparametric** tests ($p = 0.0025 < 0.05$, data not normal)

2 Exercise 2: Sign Test for One Sample

Question: Is median recovery time different from 7 days?

```
# Recovery times (days)
recovery <- c(8, 5, 7, 10, 4, 9, 3, 12, 6, 11, 7)

# Hypothesized median
med0 <- 7

# Calculate differences
diff <- recovery - med0

# Count signs (ignoring zeros)
n_positive <- sum(diff > 0)
n_total <- sum(diff != 0)

cat("Positive differences:", n_positive, "\n")
cat("Total non-zero:", n_total, "\n")

# Sign test
binom.test(n_positive, n_total, p=0.5)
```

Answers:

1. Positive differences: 5
2. Total non-zero differences: 9
3. P-value: 1.0
4. Is median significantly different from 7 days? **NO** ($p = 1.0 \geq 0.05$)

3 Exercise 3: Sign Test for Paired Data

Question: Does treatment reduce pain scores?

```
# Pain scores (1-10 scale)
before <- c(7, 8, 6, 9, 5, 8, 7, 6)
after <- c(5, 6, 4, 7, 4, 6, 5, 5)

# Calculate differences
diff <- before - after

# Count signs
n_improved <- sum(diff > 0)
n_total <- sum(diff != 0)

cat("Number improved:", n_improved, "\n")
cat("Number total:", n_total, "\n")

# Sign test
binom.test(n_improved, n_total, p=0.5)

# Summary
mean(before)
mean(after)
mean(diff)
```

Answers:

1. Number improved: 8
2. P-value: 0.0078
3. Does treatment reduce pain? **YES** ($p = 0.0078 < 0.05$)
4. Average reduction in pain score: 1.75

4 Exercise 4: Blood Pressure Study

Research Question: Does medication reduce blood pressure?

```
# Blood pressure before and after medication
bp_before <- c(145, 138, 150, 142, 156, 148, 140, 152)
bp_after <- c(138, 135, 142, 140, 148, 145, 138, 146)

# Check normality of differences
diff <- bp_before - bp_after
shapiro.test(diff)

# Wilcoxon signed-rank test
wilcox.test(bp_before, bp_after, paired=TRUE)

# Summary statistics
mean(bp_before)
mean(bp_after)
mean(diff)
```

Answers:

1. Shapiro-Wilk p-value for differences: 0.0711
Normality assumption OK? YES ($p = 0.0711 \geq 0.05$, though borderline)
2. Wilcoxon test p-value: 0.0139
3. Significant at $\alpha = 0.05$? YES ($p = 0.0139 < 0.05$)
4. Does medication reduce BP? YES ($p < 0.05$, significant reduction)
5. Mean reduction in BP: 4.88 mmHg

5 Exercise 5: Physical Therapy Study

Question: Does physical therapy reduce pain scores?

```
# Pain scores (0-10 scale) before and after 8 weeks PT
pain_before <- c(7, 8, 6, 9, 7, 8, 6, 7, 8, 9)
pain_after <- c(5, 6, 5, 7, 6, 7, 4, 6, 7, 8)

# Check normality
diff <- pain_before - pain_after
shapiro.test(diff)

# Wilcoxon test
wilcox.test(pain_before, pain_after, paired=TRUE)

# Medians
median(pain_before)
median(pain_after)
median(diff)
```

Answers:

1. Are differences normally distributed? **NO** ($p = 0.0002 < 0.05$)
2. Wilcoxon p-value: 0.0046
3. Does PT significantly reduce pain? **YES** ($p < 0.05$)
4. Median reduction in pain score: 1

6 Exercise 6: Treatment Comparison

Research Question: Do two treatments differ in recovery time?

```
# Recovery times (days)
treatment_A <- c(23, 31, 25, 28, 30, 27, 29)
treatment_B <- c(18, 22, 20, 24, 19, 21, 23)

# Check normality
shapiro.test(treatment_A)
shapiro.test(treatment_B)

# Wilcoxon rank-sum test
wilcox.test(treatment_A, treatment_B)

# Summary statistics
median(treatment_A)
median(treatment_B)
```

Answers:

1. Treatment A normal? YES ($p = 0.8525 \geq 0.05$)
Treatment B normal? YES ($p = 0.9493 \geq 0.05$)
Note: Both are normal, but we can still use Wilcoxon as a robust alternative
2. Wilcoxon rank-sum p-value: 0.004
3. Significant at $\alpha = 0.05$? YES ($p = 0.004 < 0.05$)
4. Which treatment has shorter recovery? **Treatment B** (median 21 days vs 28 days)
5. Median recovery times:
 - Treatment A: 28 days
 - Treatment B: 21 days

7 Exercise 7: Drug Effectiveness Study

Question: Do two drugs differ in effectiveness?

```
# Symptom relief scores (higher = better)
drug_X <- c(45, 52, 48, 55, 50, 47, 53)
```

```

drug_Y <- c(38, 42, 40, 45, 39, 43, 41)

# Wilcoxon test
wilcox.test(drug_X, drug_Y)

# Summary
mean(drug_X)
mean(drug_Y)
median(drug_X)
median(drug_Y)

```

Answers:

1. Wilcoxon p-value: 0.0026
2. Which drug is more effective? **Drug X** (higher scores)
3. Difference in average scores: $50 - 41.14 = 8.86$ points

8 Exercise 8: Pain Relief Comparison

Research Question: Do three pain medications differ in effectiveness?

```

# Pain relief scores (0-10 scale, higher = better)
drug_A <- c(3, 4, 5, 4, 3, 5, 4)
drug_B <- c(5, 6, 7, 6, 5, 7, 6)
drug_C <- c(7, 8, 9, 8, 7, 9, 8)

# Combine data
pain_scores <- c(drug_A, drug_B, drug_C)
drug <- factor(rep(c("A", "B", "C"), each=7))

# Kruskal-Wallis test
kruskal.test(pain_scores ~ drug)

# Post-hoc pairwise comparisons
pairwise.wilcox.test(pain_scores, drug, p.adjust.method="bonferroni")

# Summary statistics
tapply(pain_scores, drug, median)

```

Answers:

1. Kruskal-Wallis p-value: 0.0002
2. Significant overall at $\alpha = 0.05$? YES ($p = 0.0002 < 0.05$)
3. Post-hoc tests (Bonferroni-adjusted):
 - A vs B: $p = 0.0121$ (SIGNIFICANT)
 - A vs C: $p = 0.0057$ (SIGNIFICANT)

- B vs C: $p = 0.0121$ (SIGNIFICANT)
 - All three drugs differ significantly from each other
4. Most effective drug: **Drug C** (median = 8)
 5. Least effective drug: **Drug A** (median = 4)

9 Exercise 9: Comparing Parametric vs Nonparametric

Question: Sometimes data are borderline normal. Let's compare both approaches.

```
# Blood pressure data (from Exercise 4)
bp_before <- c(145, 138, 150, 142, 156, 148, 140, 152)
bp_after <- c(138, 135, 142, 140, 148, 145, 138, 146)

# Check normality
diff <- bp_before - bp_after
shapiro.test(diff)

# Parametric: Paired t-test
t_result <- t.test(bp_before, bp_after, paired=TRUE)
cat("Paired t-test p-value:", t_result$p.value, "\n")

# Nonparametric: Wilcoxon
w_result <- wilcox.test(bp_before, bp_after, paired=TRUE)
cat("Wilcoxon p-value:", w_result$p.value, "\n")

# Compare
cat("\nBoth tests significant? Compare p-values to 0.05\n")
```

Answers:

1. Is normality assumption met ($p \geq 0.05$)? **YES** (borderline, $p = 0.0711$)
2. Which test gives smaller p-value? **t-test** (0.0012 vs 0.0139)
3. Do both reach same conclusion? **YES** (both significant, $p < 0.05$)
4. When borderline normal, which is safer? **Nonparametric** (more robust)

10 Practice Problem 1: Hospital Stay Duration

```
# Hospital stay (days)
procedure_1 <- c(3, 4, 5, 3, 6, 4, 7, 5, 4, 12)
procedure_2 <- c(5, 6, 8, 7, 9, 6, 8, 10, 7, 15)

# Check normality
shapiro.test(procedure_1)
shapiro.test(procedure_2)
```

```
# Wilcoxon rank-sum test
wilcox.test(procedure_1, procedure_2)

# Summary statistics
median(procedure_1)
median(procedure_2)
```

Answers:

1. Procedure 1 normal? **NO** ($p = 0.0083 < 0.05$)
 Procedure 2 normal? **YES** ($p = 0.0538 \geq 0.05$, borderline)
 Since Procedure 1 is not normal, use nonparametric test
2. Wilcoxon rank-sum p-value: 0.0108
3. Significant difference? **YES** ($p = 0.0108 < 0.05$)
4. Shorter stays: **Procedure 1** (median 4.5 days vs 7.5 days)
 - Median Procedure 1: 4.5 days
 - Median Procedure 2: 7.5 days
5. Why is Wilcoxon appropriate? Data not normal + outliers present (12, 15 days)

11 Practice Problem 2: Diet Program Comparison

```
# Weight loss (kg) after 3 months
diet_A <- c(2.1, 3.2, 2.5, 3.0, 2.8, 2.6)
diet_B <- c(3.5, 4.2, 3.8, 4.0, 3.9, 3.7)
diet_C <- c(4.5, 5.1, 4.8, 5.3, 4.9, 5.0)
diet_D <- c(2.8, 3.0, 2.9, 3.1, 2.7, 2.9)

# Combine data
weight_loss <- c(diet_A, diet_B, diet_C, diet_D)
diet <- factor(rep(c("A", "B", "C", "D"), each=6))

# Kruskal-Wallis test
kruskal.test(weight_loss ~ diet)

# Post-hoc tests
pairwise.wilcox.test(weight_loss, diet, p.adjust.method="bonferroni")

# Summary statistics
tapply(weight_loss, diet, median)
```

Answers:

1. Kruskal-Wallis p-value: 0.000195
2. Significant overall? **YES** ($p = 0.000195 < 0.05$)

3. Post-hoc tests (Bonferroni-adjusted) - Significant pairs:

- A vs B: $p = 0.013$ (SIGNIFICANT)
- A vs C: $p = 0.013$ (SIGNIFICANT)
- A vs D: $p = 1.000$ (NOT significant)
- B vs C: $p = 0.013$ (SIGNIFICANT)
- B vs D: $p = 0.030$ (SIGNIFICANT)
- C vs D: $p = 0.030$ (SIGNIFICANT)

4. Most effective diet: **Diet C** (median = 4.95 kg)

5. Median weight loss by diet:

- Diet A: 2.7 kg
- Diet B: 3.85 kg
- Diet C: 4.95 kg
- Diet D: 2.9 kg

12 Practice Problem 3: Satisfaction Ratings

Question: Compare customer satisfaction ratings across three stores.

```
# Satisfaction ratings (1-5 scale, ordinal)
store_A <- c(3, 4, 3, 5, 4, 3, 4, 3, 4, 3)
store_B <- c(4, 5, 4, 5, 5, 4, 5, 4, 5, 4)
store_C <- c(2, 3, 2, 4, 3, 2, 3, 2, 3, 3)

# Combine
satisfaction <- c(store_A, store_B, store_C)
store <- factor(rep(c("A", "B", "C"), each=10))

# Kruskal-Wallis (appropriate for ordinal data!)
kruskal.test(satisfaction ~ store)

# Post-hoc
pairwise.wilcox.test(satisfaction, store, p.adjust.method="bonferroni")

# Medians
tapply(satisfaction, store, median)
```

Answers:

1. Why is nonparametric appropriate? Data is **ORDINAL** (1-5 rating scale)
2. Kruskal-Wallis p-value: 0.0002
3. Do stores differ in satisfaction? **YES** ($p < 0.05$)
4. Highest satisfaction: **Store B** (median = 4.5)

5. Significant pairs:

- A vs B: $p = 0.028$ (SIGNIFICANT)
- A vs C: $p = 0.042$ (SIGNIFICANT)
- B vs C: $p = 0.0007$ (SIGNIFICANT)

13 Key Interpretations

13.1 Normality Testing

- If Shapiro-Wilk $p \geq 0.05$: Data consistent with normal distribution, can use parametric tests
- If Shapiro-Wilk $p < 0.05$: Data significantly non-normal, use nonparametric tests
- Borderline cases ($p \approx 0.05$): Use nonparametric for safety

13.2 Wilcoxon Tests

- **Paired:** Use when comparing before/after measurements on same subjects
- **Rank-sum:** Use when comparing two independent groups
- Interpret p-values same as t-tests: $p < 0.05$ = significant difference

13.3 Kruskal-Wallis Test

- Tests if at least one group differs from others
- If significant ($p < 0.05$), follow up with post-hoc pairwise comparisons
- Bonferroni correction adjusts for multiple comparisons (more conservative)

13.4 Common Student Errors to Watch For

1. **Forgetting to check normality first**
2. **Using paired=TRUE for independent groups** (should only be for paired data)
3. **Interpreting medians as means** (nonparametric tests focus on medians)
4. **Not following up significant Kruskal-Wallis with post-hoc tests**
5. **Comparing p-values to 0.01 instead of 0.05** (unless specified)